6 RQ3: effect of personas on attitudes and annotations

This section investigates two related questions. First, we examine whether personas significantly impact models' attitude scores. Second, we investigate how personas' attitude scores affect downstream annotation tasks. To this end, we adapt a previous study (Sap et al., 2022) that examines the link between human annotators's attitudes and their annotations for toxic language.

6.1 Datasets

Attitude questionnaires used by Sap et al. (2022) that cover seven attitude dimensions: valuing the freedom of offensive speech (Cowan et al., 2002), perceiving the harm of hate speech (Cowan et al., 2002), endorsement of racist beliefs (McConahay, 1986), traditionalism (Jr and McGue, 2003), language purism (Sap et al., 2022), empathy (Pulos et al., 2004), and altruism (Steg et al., 2014). The original questionnaires were composed of 27 items, which may be too few to reliably measure personas' attitude scores. To improve the reliability of results, we use 31 prompt paraphrases for each item and average the returned scores. Details in Table 3 in Appendix A.

Toxicity data is composed of 626 tweets drawn by Sap et al. (2022) from toxic language detection corpora. Each text contains annotations on whether it targets black people, is written in African-American English (AAE), or includes vulgar language. The dataset also includes demographic information and attitude values for 184 annotators, with their corresponding annotations on the offensiveness and racism levels of tweets (on a Likert scale from 1 to 5). Sap et al. (2022) used the data to examine the associations between annotators' attitudes and their toxicity ratings for the three tweet categories above. In our experiments, each tweet is fed twice to each persona: once for racism annotation and once for offensiveness annotation.

Table 4 in Appendix A shows an example for each dataset.

Metrics. As attitudes scores, we report the average questionnaire response for each attitude dimension. For toxicity, we measure average offensiveness and racism ratings (to compare personas' sensitivity to toxicity), and agreement with human annotations (Krippendorf's alpha (Krippendorff, 2018, 1970)).

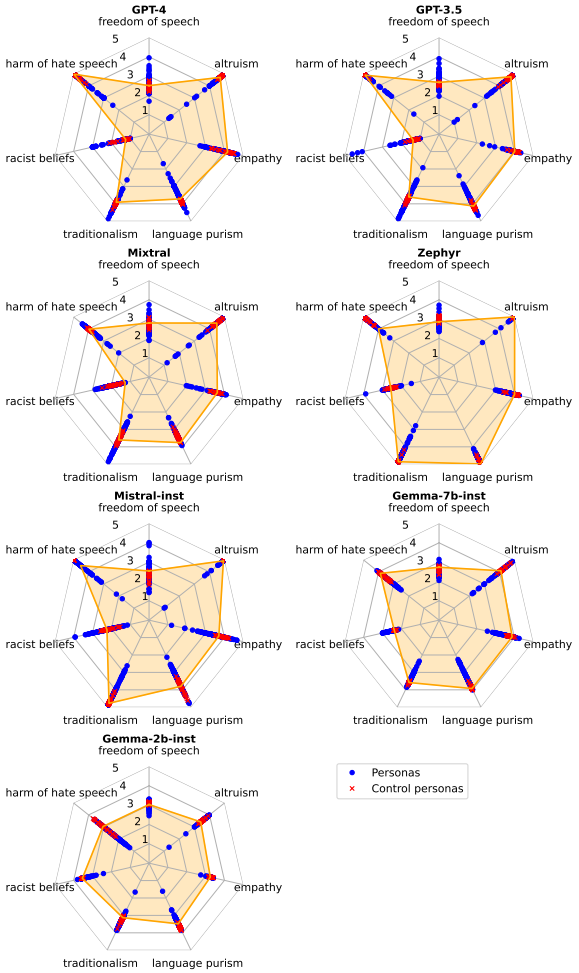

Figure 4: Distribution of attitude scores for each model. The yellow line shows the *no persona* scores.

6.2 Attitude results

Fig. 4 shows the distributions of personas' scores for each attitude and model.

Personas significantly affect attitude scores in most cases. For each model and attitude, we run a Friedman's test (Friedman, 1937) to reject the null hypothesis that personas' questionnaire responses have the same distribution.[11] Table 10 in Appendix B shows all p-values. Most results are significant for control and regular personas, but control personas have more non-significant results (14 of 49 model-attitude pairs against five for regular personas). In most cases, personas did not significantly impact freedom of speech scores. Exceptions were GPT-4 (regular personas only), Mixtral (regular only), and Mistral-inst (regular and

[11] We do not run a Cochran's test because it requires binary responses, whereas responses for the attitude and toxicity data are in a scale from 1 to 5.

control).

Regular personas have more diverse behaviors than control personas. Regular personas exhibit greater attitude score variance than control personas for all attitudes and models examined. Similarly to the previous sections, control personas are concentrated around the *no persona* baseline. However, not even regular personas could cover the full range of attitude values. For example, personas rarely exhibit high racist belief scores, in most cases exhibiting scores around 3 or less (out of 5). There are some outliers, however. For GPT-3.5, *Benito Mussolini*, *person with fascism ideology*, and *Adolf Hitler* exhibited high racist belief scores: 4.61, 4.32, and 4.08, respectively.

Some personas rankings are consistent across models. We identify consistent rankings using the procedure described in Sec. 4.

Freedom of speech: Education personas' freedom of speech scores (averaged across models) are sorted in ascending order by the education level—except that the *uneducated person* persona is on top. Further, *man* exhibited higher freedom of speech scores than all other gender personas.

Altruism: Average persona altruism scores are sorted in ascending order by their education level. Among the ideology personas, *person with fascism ideology* exhibited the lowest altruism score (1.93; the second lowest, *person with conservatism ideology*, had 3.44). In all models, the *person of atheism belief* scored lower on altruism than the religious personas.[12]

Empathy: *person with fascism ideology* had the lowest score (2.72; the second lowest, *person with nationalism ideology*, had 3.29).

Language purism: In all models, *transgender person* and *nonbinary person* scored lower on language purism than *man* and *woman*.

Traditionalism: In all models, *man* scored higher for traditionalism than the other gender personas.

Are persona's atittude associations similar to those of humans? Even though personas significantly impact attitudes, personas' attitudes may not correspond to human expectations. For example, one could expect that a persona with a high harm of hate speech score will also have a low score for racist beliefs. We explore this by comparing associations between personas' attitudes with those

[12]For Mixtral, *person of atheism belief* is tied with *person of Judaism belief* as the least altruistic.

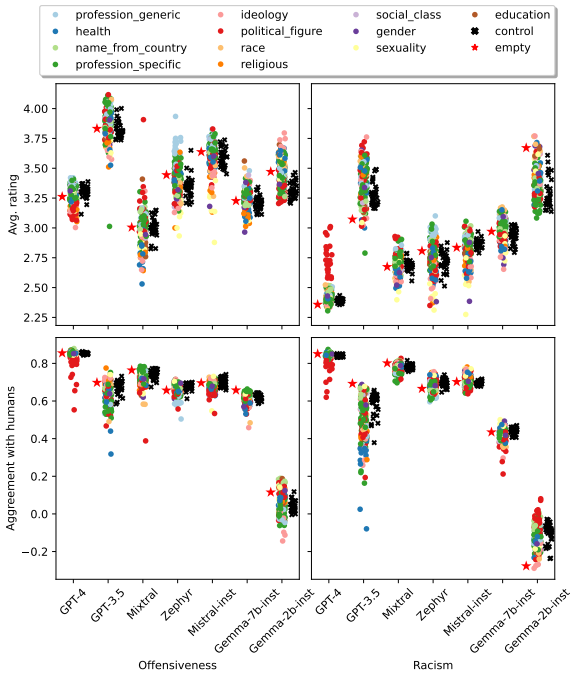

Figure 5: Distribution of toxicity scores for each model. Top row: average offensiveness and racism ratings. Bottom row: agreement with human annotations for offensiveness and racism. The ratings are in a Likert scale from 1 (not at all offensive/racist) to 5 (extremely offensive/racist).

in humans. We compute the Pearson correlations between attitude scores: of human annotators; and of personas in each model (Fig. 11 in Appendix B). We then calculate the cosine similarity between the correlations for humans and those for the personas (Fig. 6, left plot).

Except for Gemma-2b-inst (the weakest model), personas in all models have higher similarity to humans than a random baseline in which personas have randomly distributed attitude values. This result indicates that personas' attitude values somewhat mirror those present in humans. For example, for humans, there is a moderate negative correlation between altruism and racist belief, which is also present in all models (but Gemma-2b-inst).

6.3 Toxicity results

Fig. 5 shows the distributions of personas' toxicity metrics.

Personas significantly affect toxicity scores. In all cases, personas significantly impact models' answer distributions (Friedman's test, p-value $<$ .001). As in previous cases, regular personas had greater variability than control personas, which

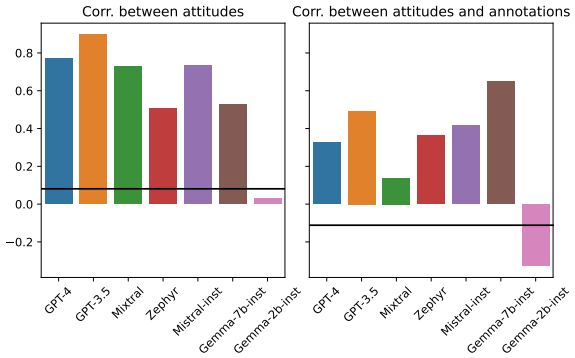

Figure 6: Cosine similarity between human and model correlations (between attitudes on the left and between attitudes and annotations on the right). The black horizontal line denotes the cosine similarity between human and random baseline correlations.

tended to concentrate around the *no persona* baseline—with some exceptions. For example, GPT-3.5 control personas rated the tweets as more racist than the *no persona* baseline and also had lower human agreement than the baseline. An interesting outlier for GPT-3.5 was the *comedian* persona, which labeled the tweets as having much lower offensive and racist content than all the other personas did.

How similar are the associations between personas' attitudes and their toxicity ratings to those of human annotators? Even though personas' attitude associations are similar to human annotators', associations between attitudes and toxicity ratings may differ for humans and personas. For example, one could expect that a persona with a high harm of hate speech score will also annotate tweets targeting black people as having higher racism and offensiveness scores. To investigate this, we compute the Pearson correlations between attitude scores and the average offensiveness and racism ratings given to three subsets of tweets: tweets in African-American English (AAE), tweets that target black people, and tweets with vulgar language. Fig. 12 in Appendix B shows the obtained correlations. Fig. 6 (right plot) shows the cosine similarity between humans and personas in each model.

Personas' correlations in all models but Gemma-2b-inst had greater cosine similarity with human correlations than the random baseline. The result indicates that not only do personas' attitude associations relate to those of humans but also their attitudes-annotation associations are similar to those of humans (at least for humans represented in the data). For example, for both humans and personas, harm of hate speech has a positive association with higher offensiveness and racism ratings for tweets targeting black people (except for Mixtral and Gemma-2b-inst personas).

However, persona behavior is less nuanced than those of humans. For example, the racist beliefs attitude in humans has a negative association with offensiveness scores for tweets targeting black people and a positive association with offensiveness scores for AAE tweets—which reflects annotators' racism. On the other hand, personas' associations generally do not distinguish AAE tweets from those targeting black people. An exception was the Gemma-7b-inst personas, whose association between racist belief and offensiveness scores reflect those of humans. Gemma-7b-inst was also the model with highest similarity to humans' attitude-annotation correlations.

7 RQ4: analysis of persona refusal

Models occasionally refuse to follow persona-assigned prompts by expressing either an inability to perform the task (e.g., *I'm sorry, but I can't provide personal opinions or preferences*), an inability to adopt the persona (*e.g., I cannot be a gay person, as I am an artificial intelligence and do not have a gender or personal experiences*), or outputting a blanket refusal (e.g., *I'm sorry, but I can't assist with this request*). The disparity of refusal rates across personas has implications for fairness (e.g., if personas from different demographic groups are treated differently) and may reveal models' underlying social biases.

This section examines how refusal rates differ across personas. We use regex patterns (code excerpt 3 in Appendix A) to identify model refusals. We then compute the refusal frequency for each model-persona pair in each dataset.

7.1 Results

Fig. 7 shows average (across datasets) refusal rates for all personas and models. Fig. 9 in Appendix B shows refusal rates for each dataset. Personas significantly impact (Cochran's Q test, p-value $< .001$) the refusal rates for almost all models and datasets. The only exception is Zephyr, for which personas' impact on refusals for the racism annotation task (Sec. 6) was not significant